

Module 2. Data and Measurements

- ☐ Data Types
- ☐ Statistics of Data
- ☐ Similarity and Distance Measures
- ☐ Summary

Types of Data Sets: (1) Record Data

- ❑ Relational records
 - ❑ Relational tables, highly structured
- ❑ Data matrix, e.g., numerical matrix, crosstabs

	China	England	France	Japan	USA	Total
Active Outdoors Crochet Glove		12.00	4.00	1.00	240.00	257.00
Active Outdoors Lycra Glove		10.00	6.00		323.00	339.00
InFlux Crochet Glove	3.00	6.00	8.00		132.00	149.00
InFlux Lycra Glove		2.00			143.00	145.00
Triumph Pro Helmet	3.00	1.00	7.00		333.00	344.00
Triumph Vertigo Helmet		3.00	22.00		474.00	499.00
Xtreme Adult Helmet	8.00	8.00	7.00	2.00	251.00	276.00
Xtreme Youth Helmet		1.00			76.00	77.00
Total	14.00	43.00	54.00	3.00	1,972.00	2,086.00

Image Source: http://www.ibm.com/support/knowledgecenter/SS4JCV_7.5.5/com.businessobjects.integration.eclipse.designer.doc/html/images/xtab5.gif

Person:

Pers_ID	Surname	First_Name	City
0	Miller	Paul	London
1	Ortega	Alvaro	Valencia
2	Huber	Urs	Zurich
3	Blanc	Gaston	Paris
4	Bertolini	Fabrizio	Rom

Car:

Car_ID	Model	Year	Value	Pers_ID
101	Bentley	1973	100000	0
102	Rolls Royce	1965	330000	0
103	Peugeot	1993	500	3
104	Ferrari	2005	150000	4
105	Renault	1998	2000	3
106	Renault	2001	7000	3
107	Smart	1999	2000	2

Image Source: <http://2.bp.blogspot.com/-HXyE2OrQHKo/Utp0EdlSh7I/AAAAAAAAAK8/fEOu1tXGZEq/s1600/hvbugogvpujij%5D.png>

- ❑ Transaction data

TID	Items
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

	team	coach	pla y	ball	score	game	wi n	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

- ❑ Document data: Term-frequency vector (matrix) of text documents

Types of Data Sets: (2) Graphs and Networks

- ❑ Transportation network
- ❑ World Wide Web

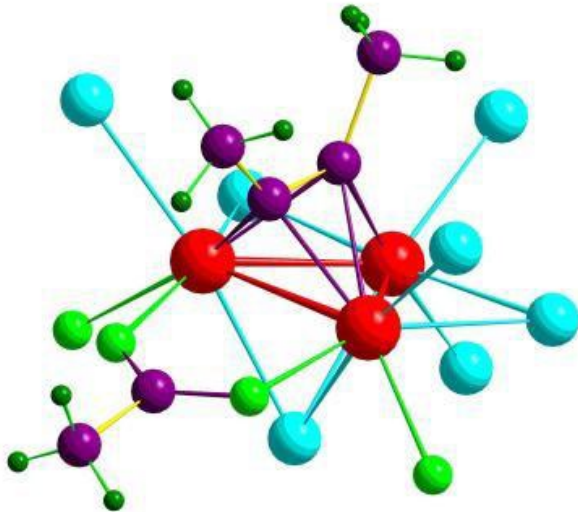


Image Source: <http://chem.ch.huji.ac.il/pics/bino.jpg>

- ❑ Molecular Structures
- ❑ Social or information networks

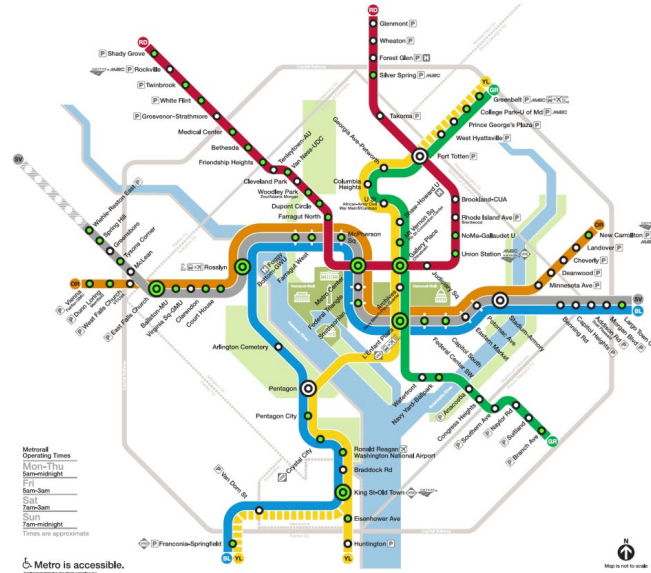


Image Source: <http://www.hotelsneardcmetro.com/wp-content/uploads/2013/03/washington-dc-metro-station-with-hotels-close-by.png>

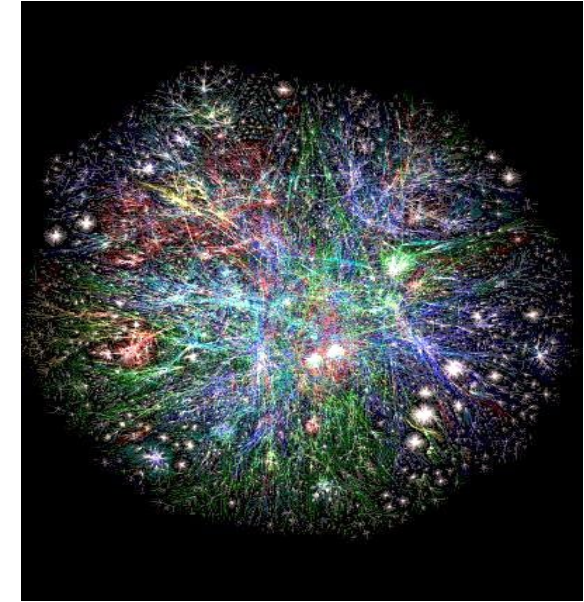


Image Source: <http://www.vlib.us/web/opte.org.jpg>

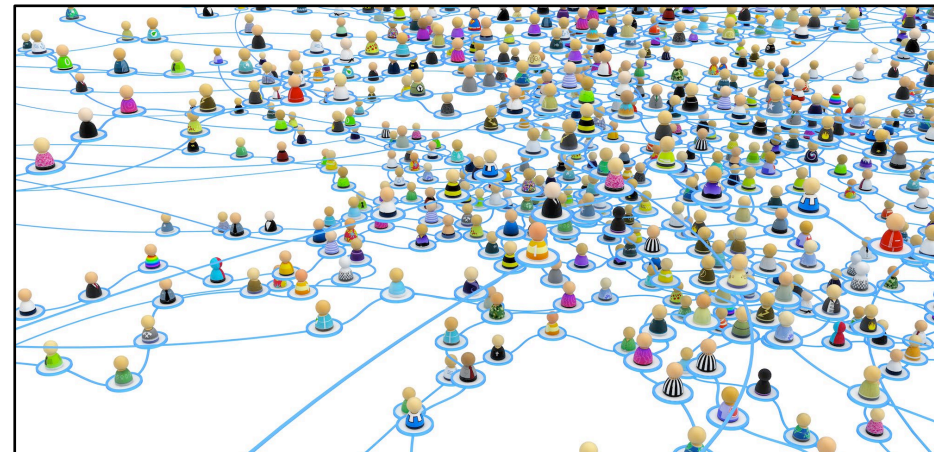


Image Source: <http://cdn1.tnwcnd.com/wp-content/blogs.dir/1/files/2013/11/social-network-links.jpg>

Types of Data Sets: (3) Ordered Data

❑ Video data: sequence of images

Image Source: <http://www.albertbelle.net/pictures/jimart/belleartbb.jpg>

❑ Temporal data: time-series

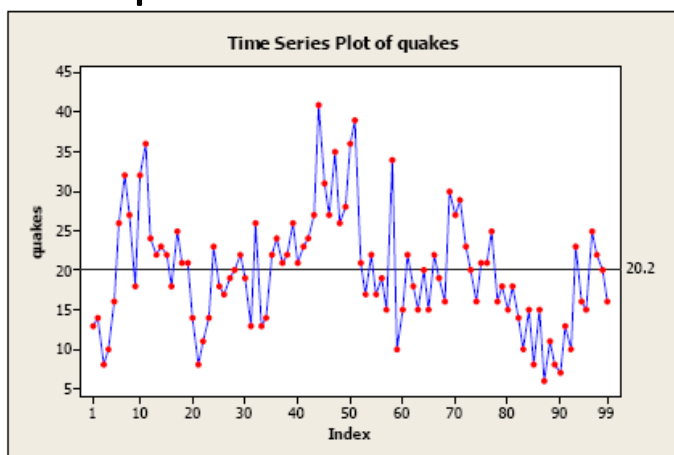
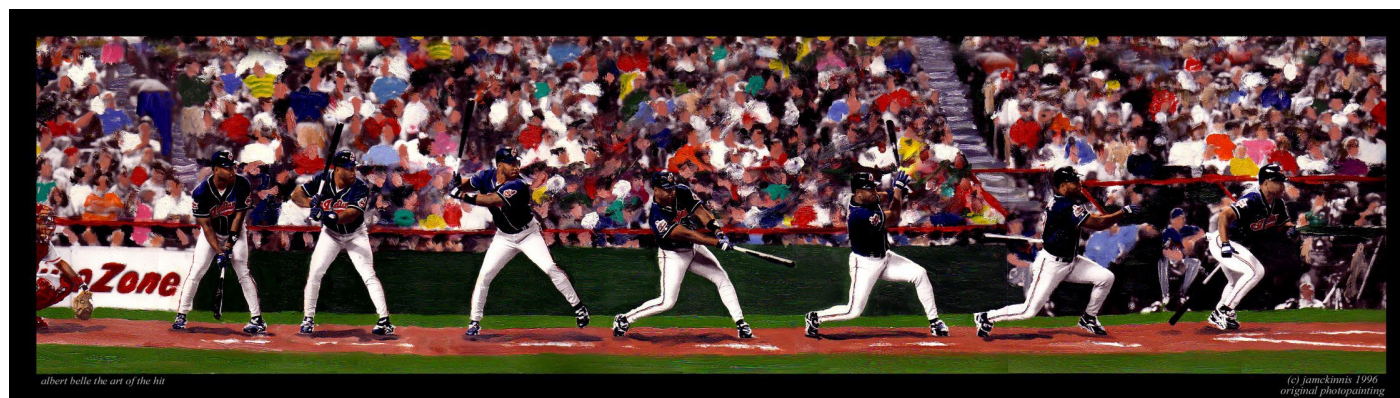


Image Source: https://onlinecourses.science.psu.edu/stat510/sites/onlinecourses.science.psu.edu.stat510/files/L01/graph_01.gif

❑ Sequential Data: transaction sequences

❑ Genetic sequence data



	Start
Human	GTTTGGAGG -- ATGTTCAACAAATGCTCCTTTTCATTCTCTATTACAGACCTGCCGCA
Chimpanzee	GTTTGGAGG -- ATGTTCAATAAATGCTGCTTTTCATTCTCTATTACAGACCTGCCGCA
Macaque	GTTTGGAGG -- ATGTTCAATAAATGCTCCTTTTCATTCTCTATTACAAACTTGCCGCA
Human	GACAATTCTGCTAGCAGCCTTTGTGCTATTATCTGTTTTCTAAACTTAGTAATTGAGTGT
Chimpanzee	GACAATTCTGCTAGCAGCCTTTGTGCTATTATCTGTTTTCTAAACTTAGTAATTGAGTGT
Macaque	GACAATTCTGCTAGCAGCCTTTGTGCTATTATCTGTTTTCTAAACTTAGTAATTGAGTGT
	↓
Human	GATCTGGAGACTAA - CTCTGAAATAAATAAGCTGATTATTTATTTATTTTCTCAAAACAA
Chimpanzee	GATCTGGAGACTAAACTCTGAAATAAATAAAGCTGATTATTTATTTATTTTCTCAAAACAA
Macaque	TATCTGGAGACTAAACTCTGAAATAAATAAAGCTGATTATTTATTTATTTTCTCAAAACAA
Human	CAGAATACGATTTAGCAAATTACTTCTTAAGATATTATTTTACATTTCTATATTCTCCTA
Chimpanzee	CAGAATACGATTTAGCAAATTACTTCTTAAGATATTATTTTACATTTCTATATTCTCCTA
Macaque	CAGAATATGATTTAGCAAATTACTTCTTAAGATATTATTTTGCACITCTATATTCTCCTA
Human	CCCTGAGTTGATGTGTGAGCAATATGTCACCTTTTCATAAAGCCAGGTATACA --- TTATG
Chimpanzee	CCCTGAGTTGATGTGTGAGCCGATATGTCACCTTTTCATAAAGCCAGGTATACA --- TTATG
Macaque	CCCTGAGTTGATGTGTGAGCAATATGTCACCTTTTCATAAAGCCAGGTATATATACATTACG
	H I I Y S T F L S K
Human	GACAGGTAAGTAAAAAACATATTATTTATTCTACGTTTTGTCCAAAAATTTTAAATTTT
Chimpanzee	GACAGGTAAGTAAAAAACATATTATTTATTCTACGTTTTGTCCAAAAATTTTAAATTTT
Macaque	GACAGGTAAGTAAAAAACATATTATTTATTCTAGGTTTTGTCCAAAGATTTTAAATTTT
Human	AACTGTTGCGCGTGTGTTGGTAA --- TGTAAAACAAAC TCAGTACA
Chimpanzee	AACTGTTGCGCGTGTGTTGGTAA --- TGTAAAACAAAC TCAGTACA
Macaque	AACTGTTGTGATGTGTTGGTAA --- CGTAAAACAAATTCAGTACG

Image Source: <http://www.panspermia.org/cllu2.jpg>

Types of Data Sets: (4) Spatial, Image, and Multimedia Data

❑ Spatial data: maps



Image Source: http://nda.membershipsoftware.org/files/iStock_000032315954_Large.jpg

❑ Image data:

❑ Video data:

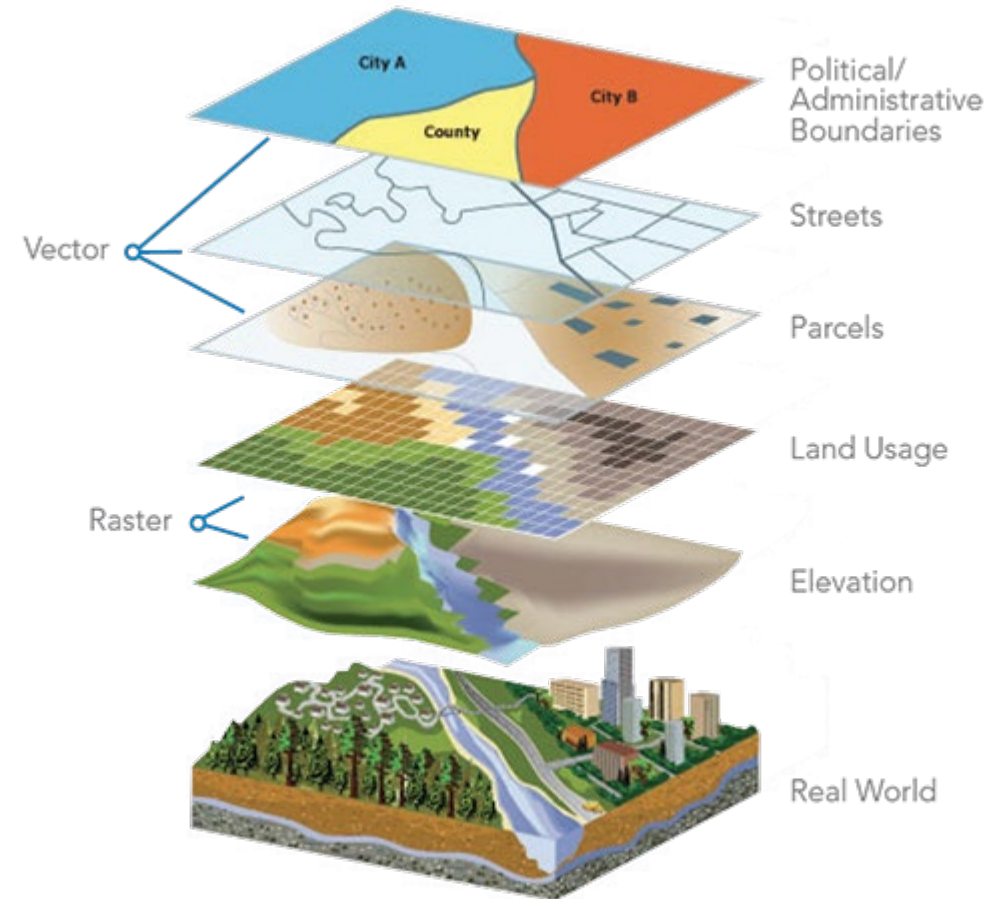


Image Source: <https://learn.arcgis.com/en/arcgis-book/images/ch5/05-fig-5-4.png>